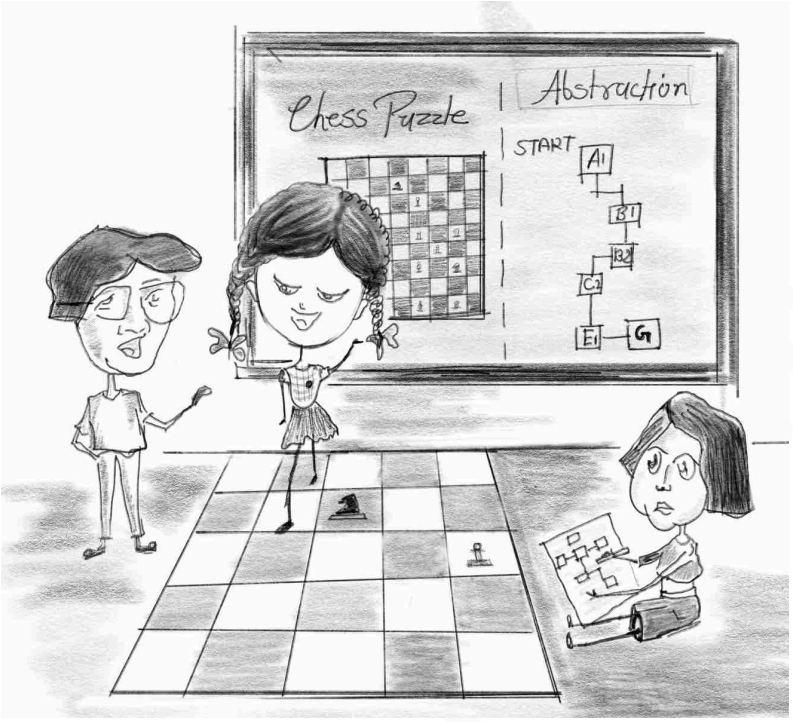
Chess Maze and Abstraction

Grade 7



Using a chess-maze puzzle, one can explore the idea of abstraction in computational thinking. Abstraction means reducing a complex problem to its essential parts, making it easier to understand and solve. In this lesson, one begins with a puzzle with chess pieces drawn on a grid. The objective is to reach the Goal by navigating using the chess rules. Explore how to represent the puzzle in a simpler form using a tree diagram. This helps one to shift from a visual puzzle to a computational activity.

Note for Teachers: Learning Goals

- Exploration: Chess maze by chess-based movement rules.
- Abstraction: keeping the important details of a puzzle only.
- Spatial Representation: Square in the puzzle as circle and legal moves as line.

This lesson can be done in one 45-minute class or split across two shorter sessions.

- Opening conversation, Introduce the puzzle and Try out puzzle: 15 minutes
- Introduction of the Abstraction and building the tree diagram: 25 minutes
- Discussion, Conclusion and Reflection: 5 minutes

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Chess Maze

Reach the Goal position by navigating using the chess rules.
Start from the top-left square (Start) and reach the bottom-right square (Goal).

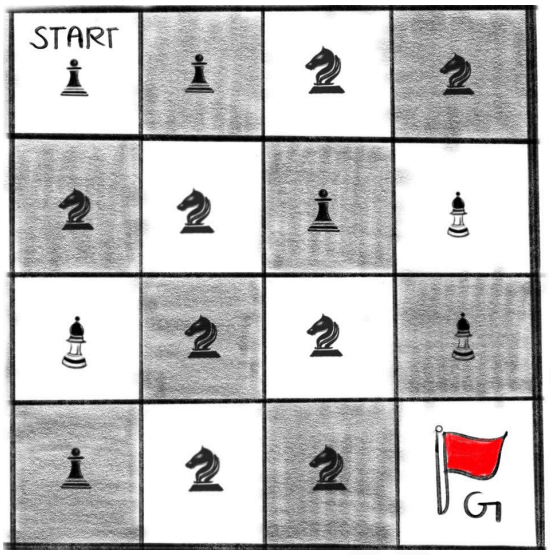


Figure: Reach the Goal

- These puzzles use the movements of the chess pieces to define how one can move from each square.
- Pawn moves one position in any direction (up, down, right, or left) within the grid.
- Bishop moves diagonally in any position within the grid.
- Knight moves in an L-shape (two spaces in one direction, then one space perpendicular) within the grid.

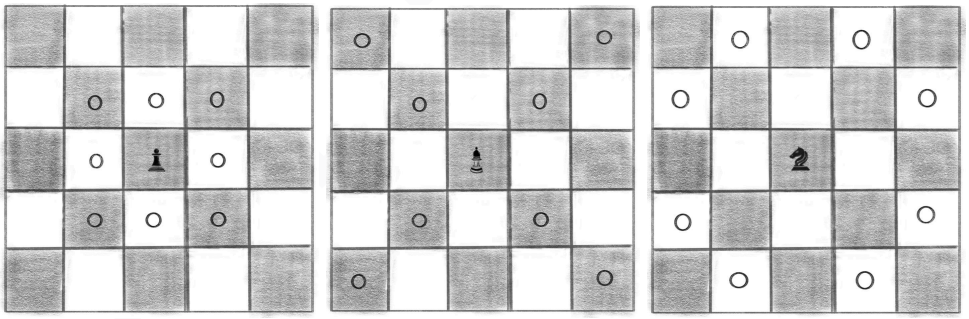


Figure: Possible movement of Pawn, Bishop and Knight

- Consider the example shown below. The 4x4 grid has one chess piece in each square. The starting position (top-left square) has the pawn as a chess piece; it can move one position in any direction (up, down, right, or left). Here, since the only option is downwards or towards the right, one can move there.

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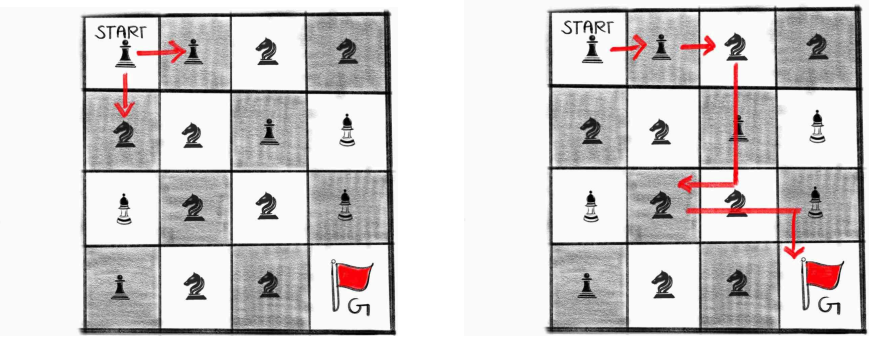


Figure: Example of a pawn move, Solution of the Puzzle

- **Not allowed**
  Pawn - move diagonally.
  The pawn can move up, down, right, or left, but not diagonally.

  Bishop - move straight.
  The bishop can move diagonally but not straight.
  The bishop cannot change the direction of steps in the middle.

  Knight - move straight or diagonal.
  The knight can move in an L-shape only but not diagonally or straight.

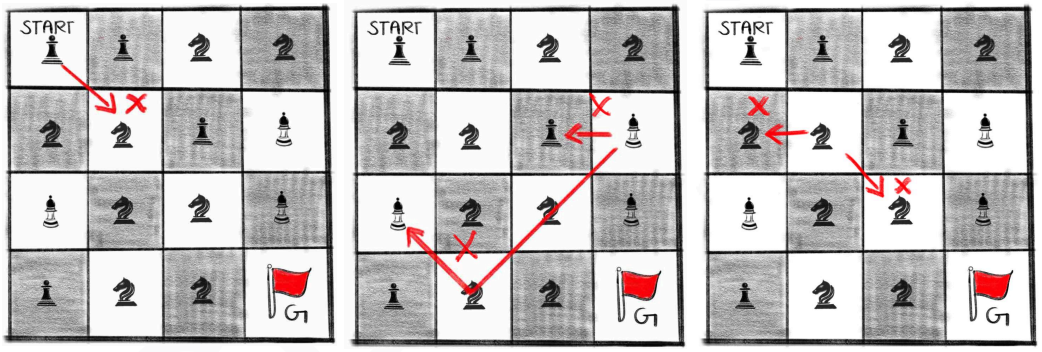


Figure: Moves Not Allowed

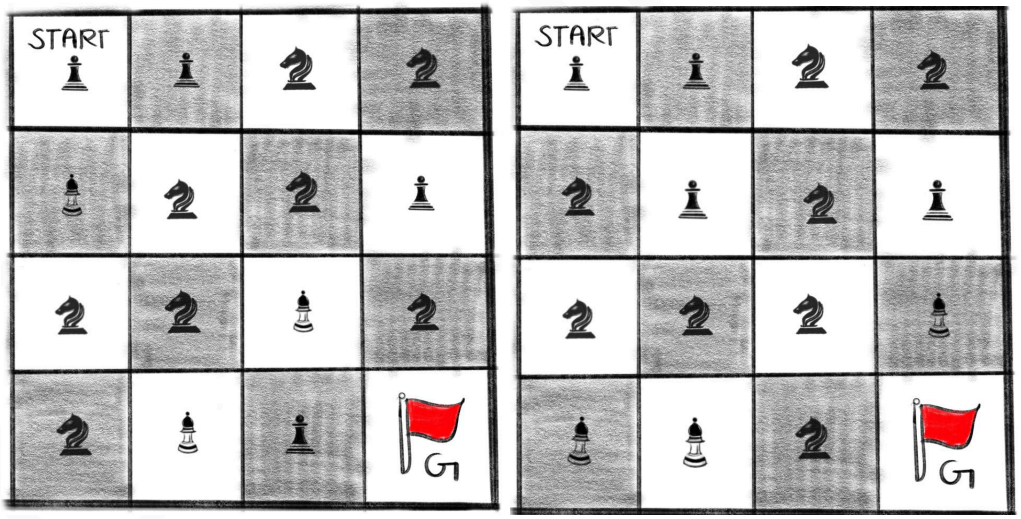
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Note for Teachers: Classroom Set-Up

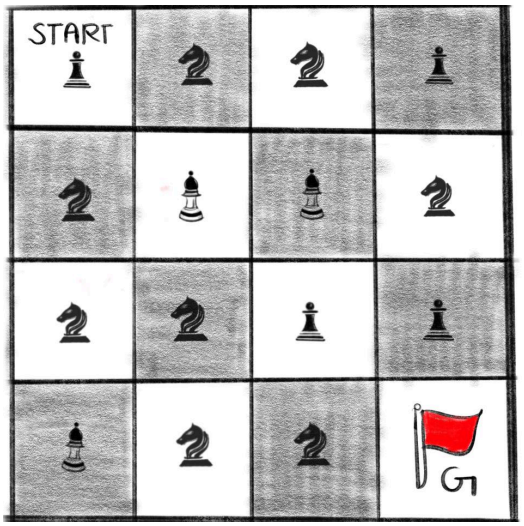
- Explain the puzzle with an demonstration of given example.
- Draw the chess maze shown as an example on the board or draw it on a floor using chalk. The square tiles are great as they can serve as a grid. Invite students to try out the solutions by jumping over the grid.
- There may be several paths to the goal. The task is to identify the shortest path among them.
- Once the puzzle is solved, More puzzle can be given for pen-paper exploration.
- Let students experience the puzzle first and notice that some positions offer several choices while others lead to dead ends. After this section, one can transition into the abstraction part of the chess maze.
- To try another example, one can visit this website to generate more such chess maze: https://chess-puzzle-architect.vercel.app/

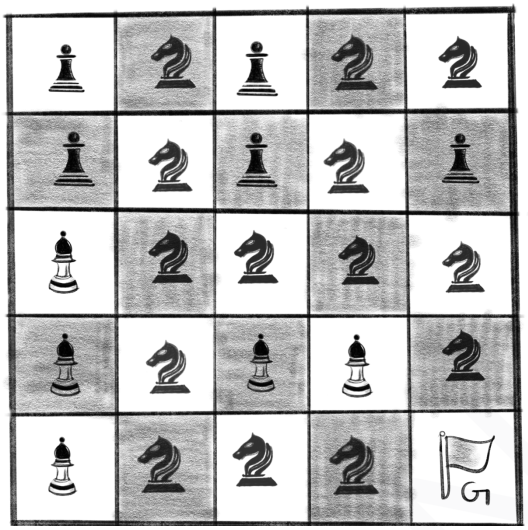
Exploration

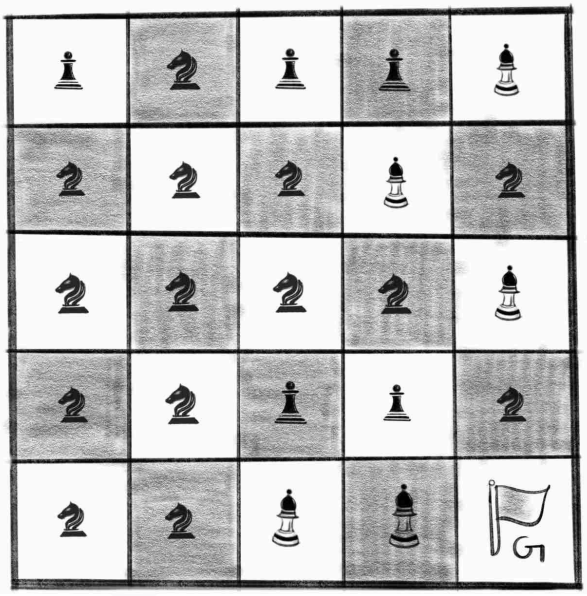
Chess path puzzle exercises to try in the classroom.

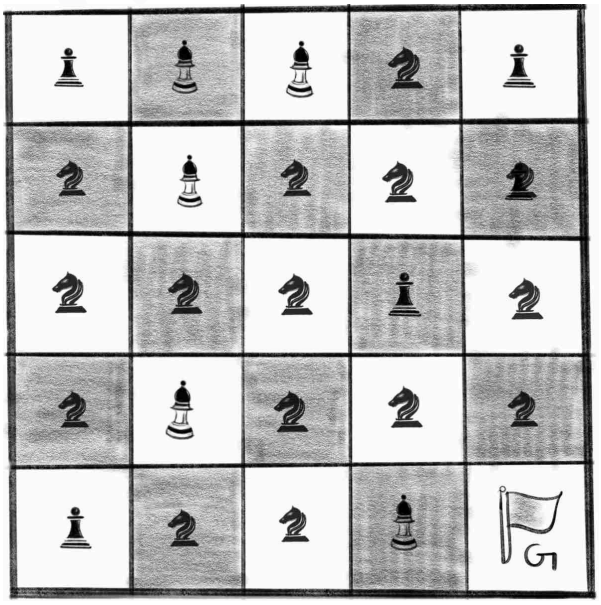


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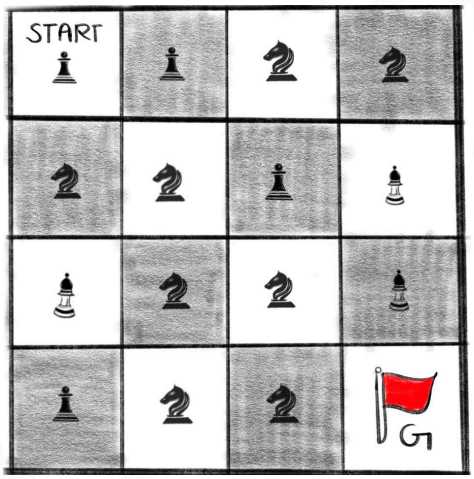
Note for Teachers: Intro to Abstraction

- If we wanted to explain this puzzle to a computer, what would we really need to include?
- Notice that the exact grid drawing style is not important. The important parts are the squares and the legal moves between them, which is the essence of abstraction.

Chess Maze: Abstraction

A computer does not "see" the number maze the way humans do. Instead, it represents the maze as a tree diagram with squares and lines. Each square remains a square, and each possible jump or number of steps becomes a connection (line) between the squares. A tree diagram is a useful way to represent connections between the squares. This process of simplifying a problem into an easier representation is called Abstraction.

Let us take the initial chess maze as an example to understand the tree diagram.



Label the squares in a maze with names such as A1, A2, A3, ... for the first row, B1, B2, B3,... for the second row, and so on. Now we should draw one line for each square and connect each square when a legal move is possible between them.

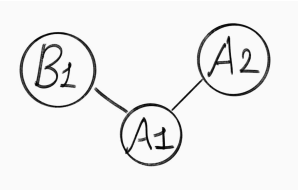
A_1	A_2	A_3	A_4
B_1	B_2	B_3	B_4
C_1	C_2	C_3	C_4
D_1	D_2	D_3	D_4

Now we need to connect the squares. First, we will take our starting square, A1. Since A1 is represented as a pawn, it should be connected to A2 and B1, since both are one step away vertically/horizontally from A1.

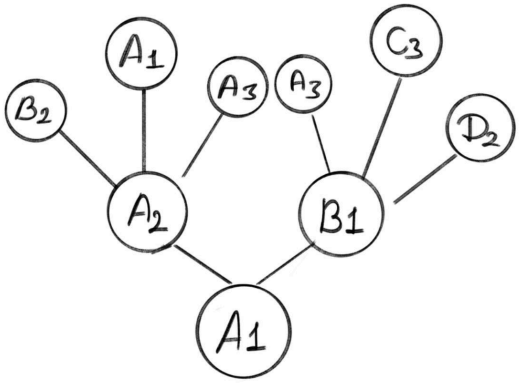
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We will start to draw the tree diagram from the start square 'A1'and model the first few branches.

Representation of A1 → A2, B1 is shown below.

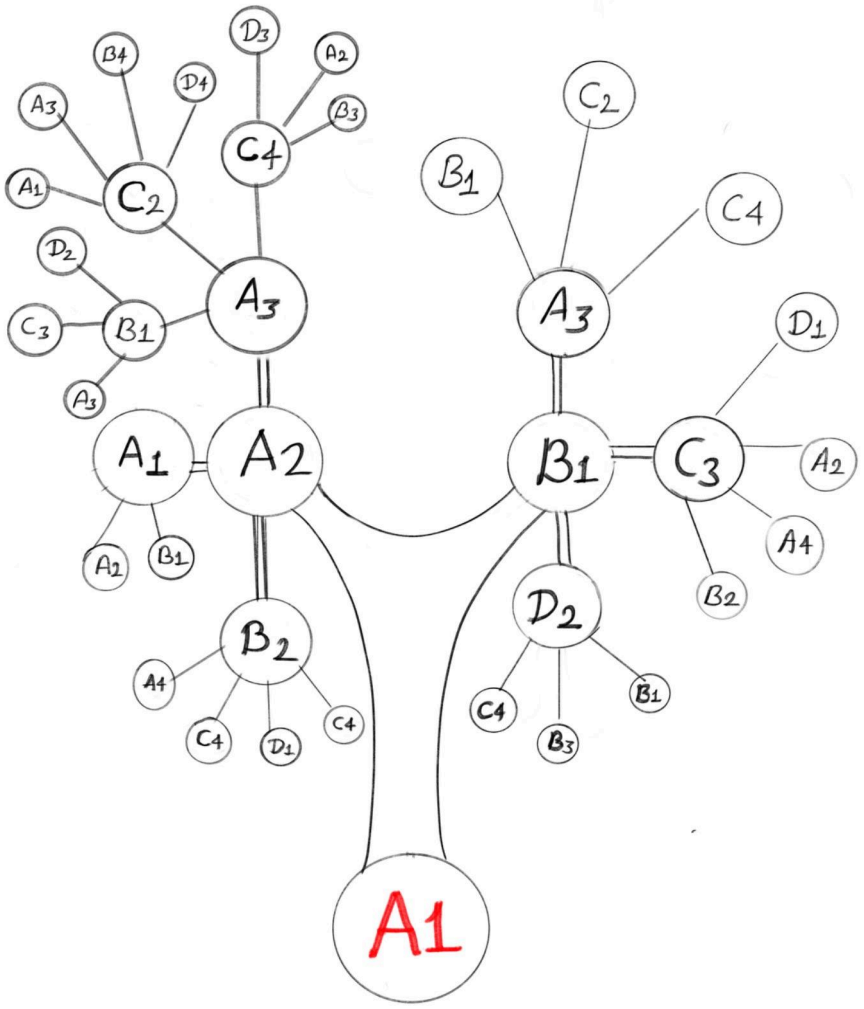


We can do this exercise for every square. Now, for Square A2 and B1, we will branch out onwards with their connections A2 → A1, A3,B2 and B1 → A3,C3,D2.



Continue drawing the tree diagram onwards here. All branching will stop once we reach the end square 'D4'.

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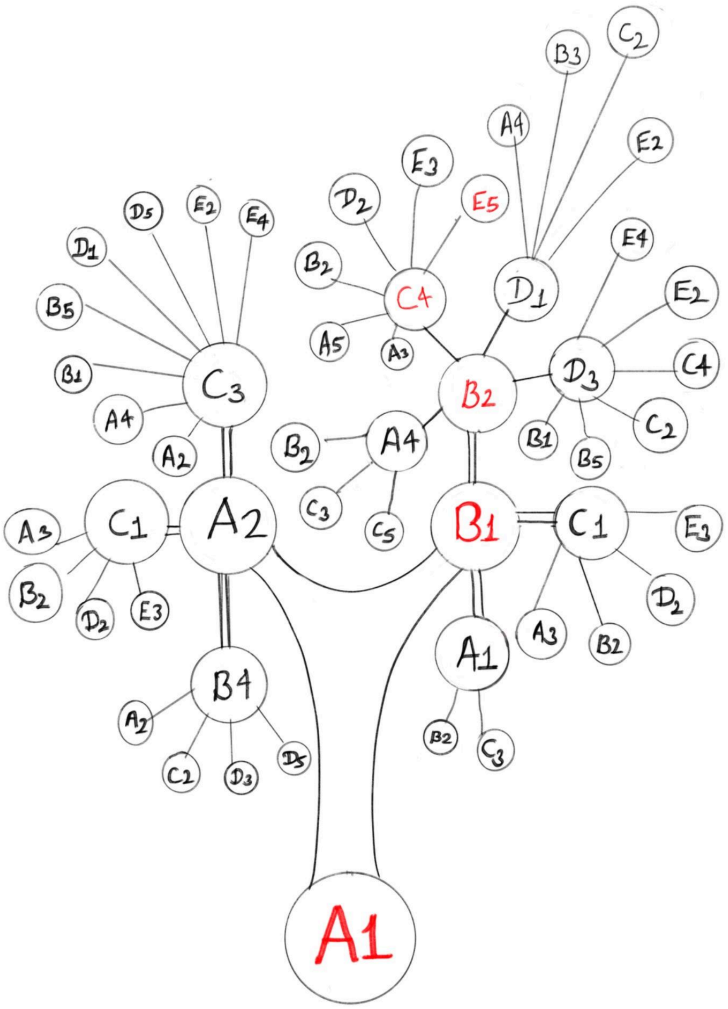


Now, instead of finding the path by moving as a chess piece, one can visually find a path from A1 to D4 from the tree diagram.

The solution path of the chess maze is shown below.

A1 (Pawn)	A2 (Pawn)	A3 (Knight)	C2 (Knight)	D4 (Goal)

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It can be shown that a complex-looking maze can be represented in a simpler, more useful way. By moving from a chess maze to a tree diagram, one experienced abstraction as an action rather than just a definition. One may discover an alternative route by exploring further, but the key is to choose the path that reaches the goal in the fewest steps.

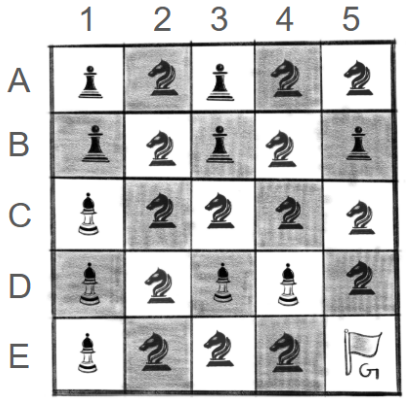
An interesting exercise could be to find the total number of paths possible to reach the goal.

Note for Teachers

- Abstraction is a central pillar of computational thinking and helps learners manage complexity by focusing on the information that matters for solving a problem. In this activity, the full grid drawing is not the only thing one need. What matters more is the set of possible positions and the allowed movements between them.
- A tree diagram is a structure of a hierarchical arrangement of with a squares and lines, where square represent objects or positions and lines represent connections between them.
- One do not need formal graph theory vocabulary at all. The lesson should start with the puzzle experience, then move to naming the simplified representation as a tree diagram.

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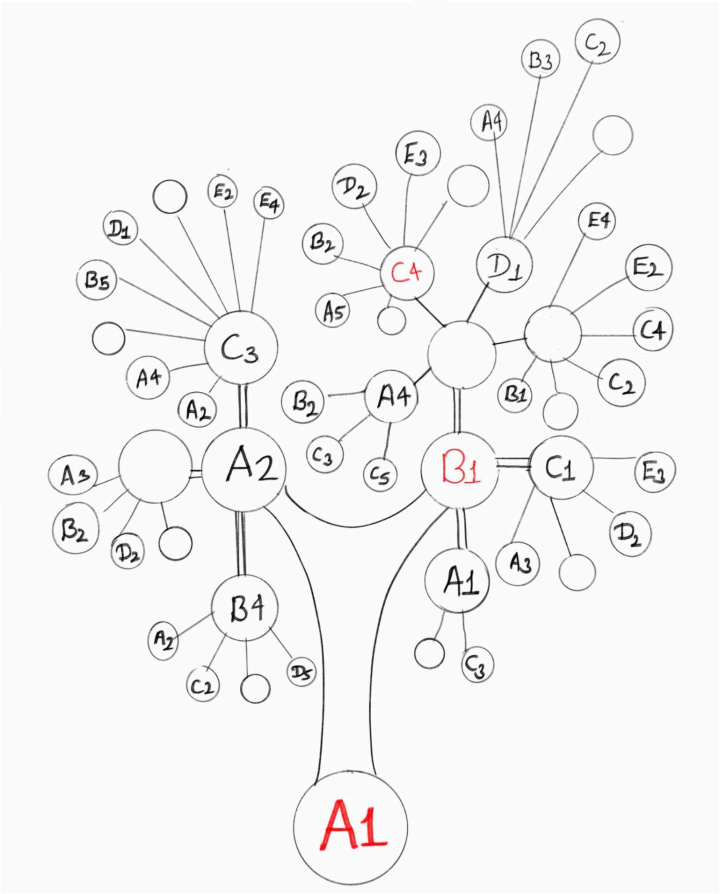
Q. For the Given Chess Maze, Answer the following Questions.



1. Which Chess piece is associated with square D2?
   a) Pawn
   b) Bishop
   c) Knight
   d) Goal

2. Which square cannot be visited from square B4(Knight)?
   a) A2 (Knight)
   b) C2 (Knight)
   c) B3 (Pawn)
   d) D5 (Bishop)

3. The tree diagram is shown below. Fill in the blank squares.

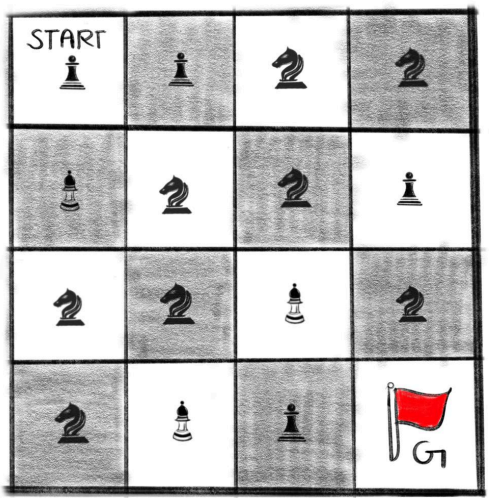


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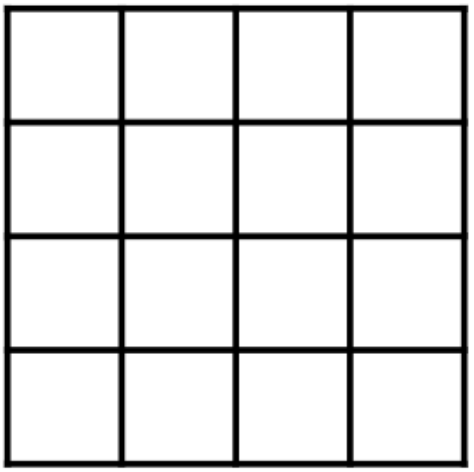
Explore More

Draw the diagram representation for the chess maze, as shown in the Puzzle Try Out examples.

A)



Label the maze square

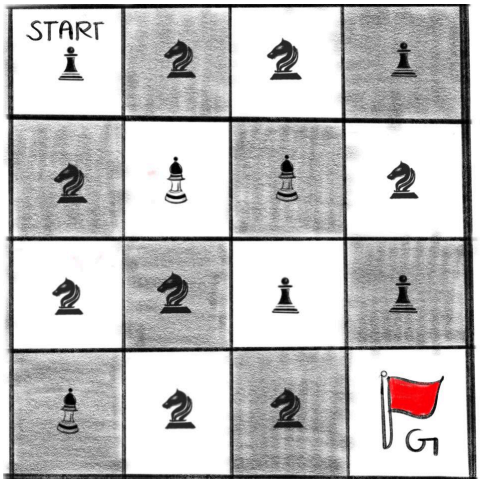


Draw the tree diagram of the maze

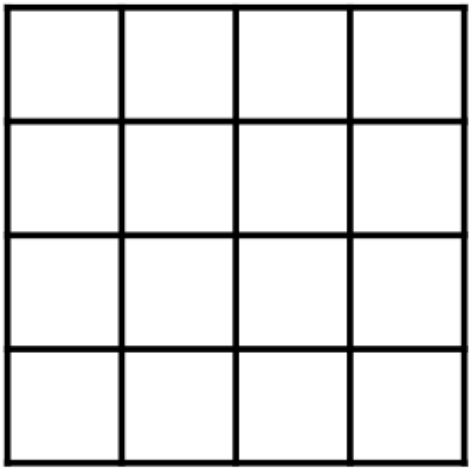
Solution path

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B)



Label the maze square

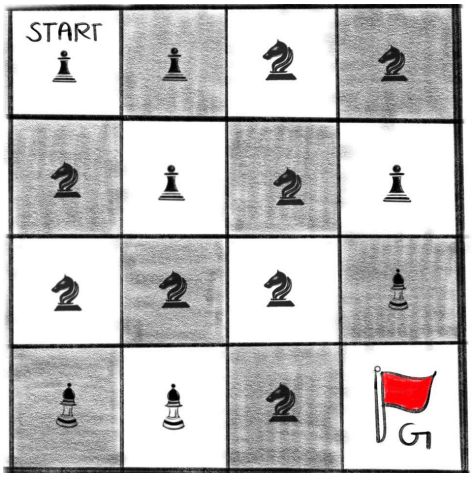


Draw the tree diagram of the maze

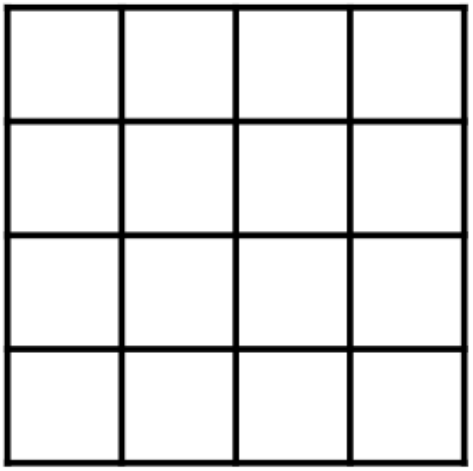
Solution path

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C)



Label the maze square

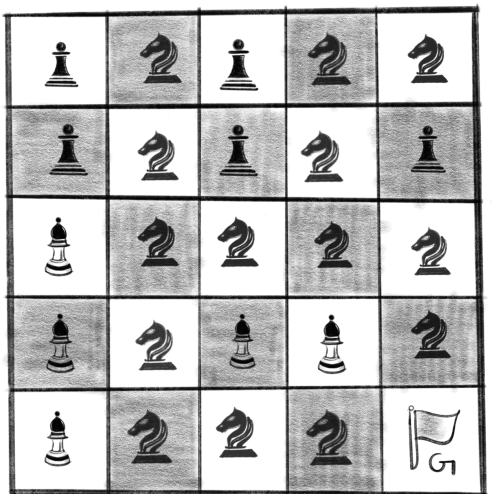


Draw the tree diagram of the maze

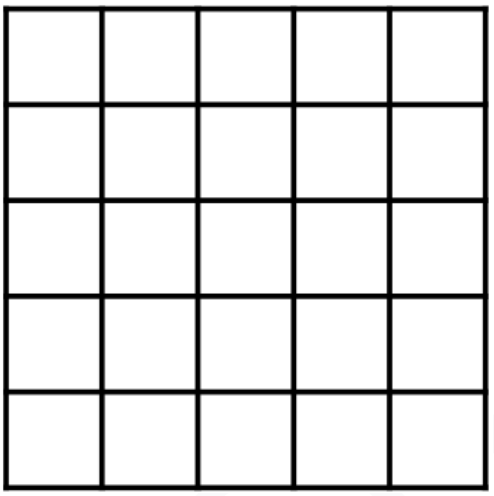
Solution path

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D)

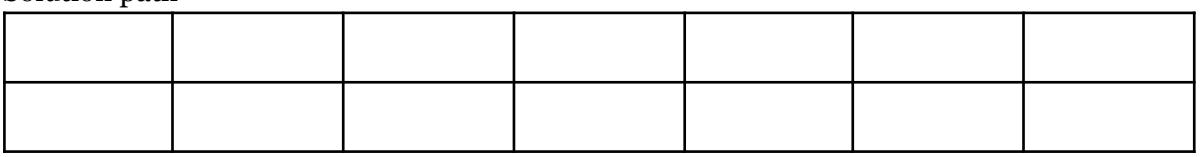


Label the maze square



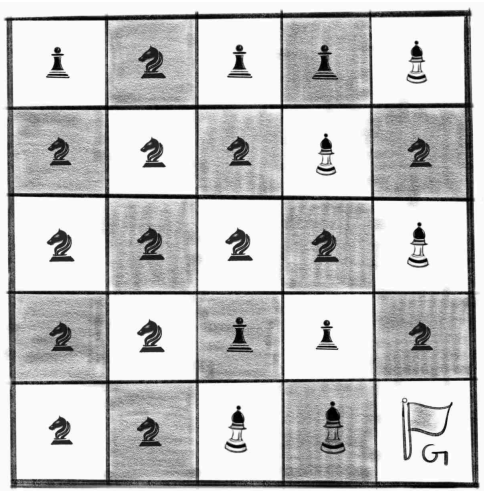
Draw the tree diagram of the maze

Solution path



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E)



Label the maze square

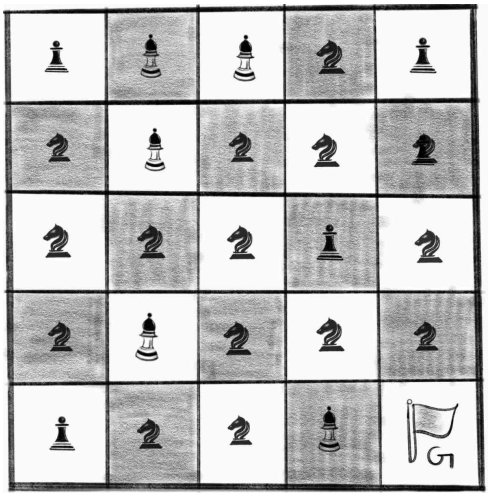
Draw the tree diagram of the maze

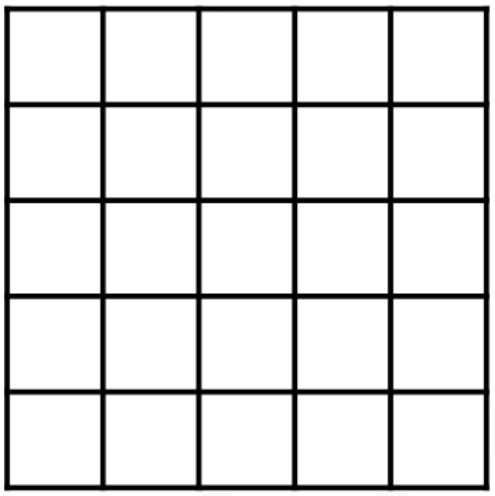
Solution path

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F)

Label the maze square





Draw the tree diagram of the maze

Solution path

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Solutions

A.

B.

C.

D.

E.

F.

Discussion, Conclusion and Reflection

The key goal of this lesson is not just puzzle-solving. It helps to understand that a complex-looking maze puzzle can be represented in simpler, more useful ways. By moving from a chess grid to a tree diagram, one has experienced abstraction as an action rather than just a definition.

Note for Teachers : Discussion, Conclusion and Reflection

- Bring the class together and compare the answers.
- Use the following discussion questions:
  What information from the puzzle was important to keep?
  What information could be ignored?
- This conversation helps to see that abstraction is not about removing meaning. It is about maintaining the structure needed to solve the problem.

Here, using the tree diagram, we can visually see the maze's path, whereas a computer does not find the path visually; it needs a strategy. We will learn about different strategy computer used to find the solution path later on.

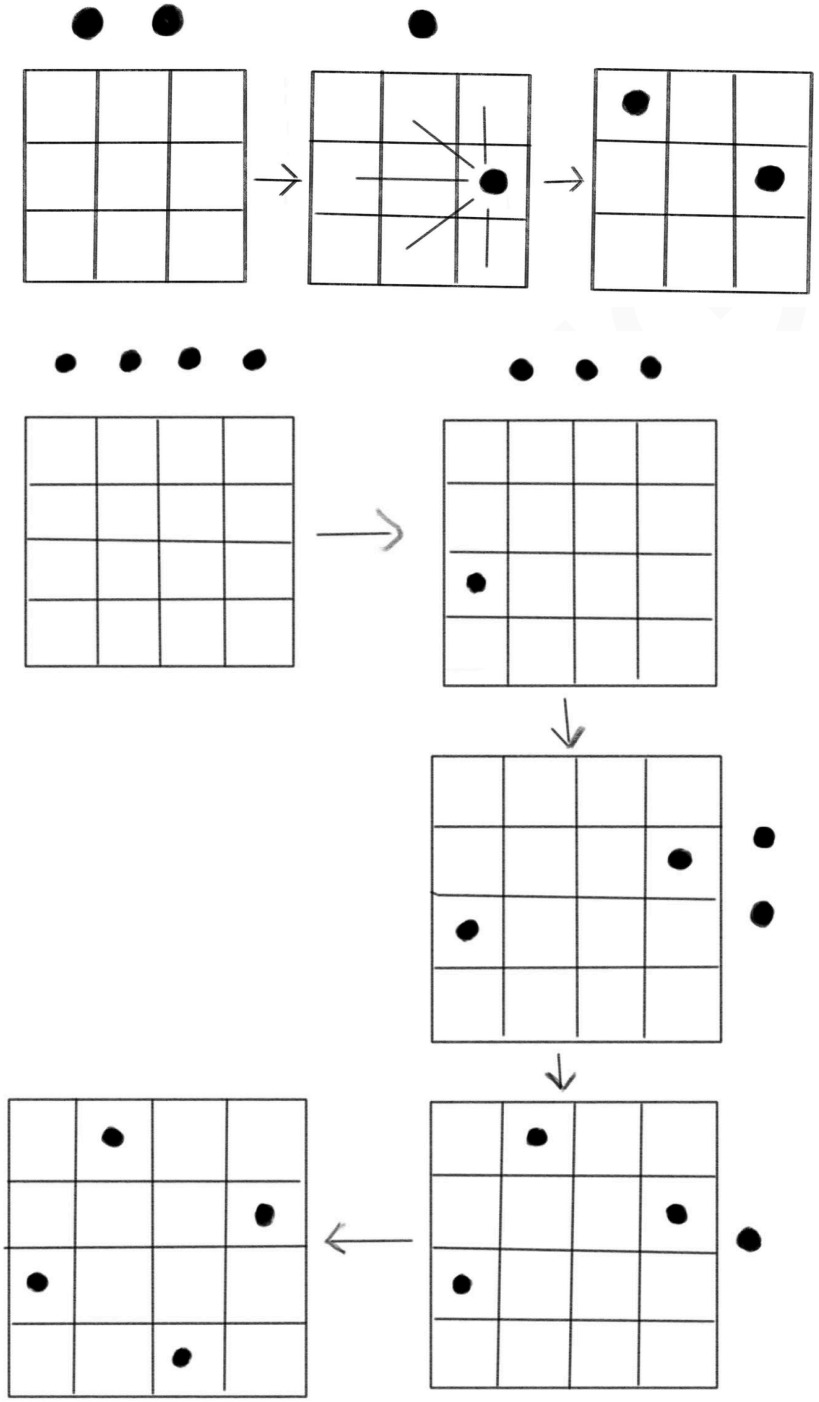
A tree diagram lets us quickly find the maze's solution path by showing all possible moves and where paths split. One can easily look over the branches to see which routes lead to the goal. Computers don't "look" at paths the way we do; they use specific rules and step-by-step strategies to explore options. In later lessons, we'll study those search methods and watch how they reliably locate a solution.

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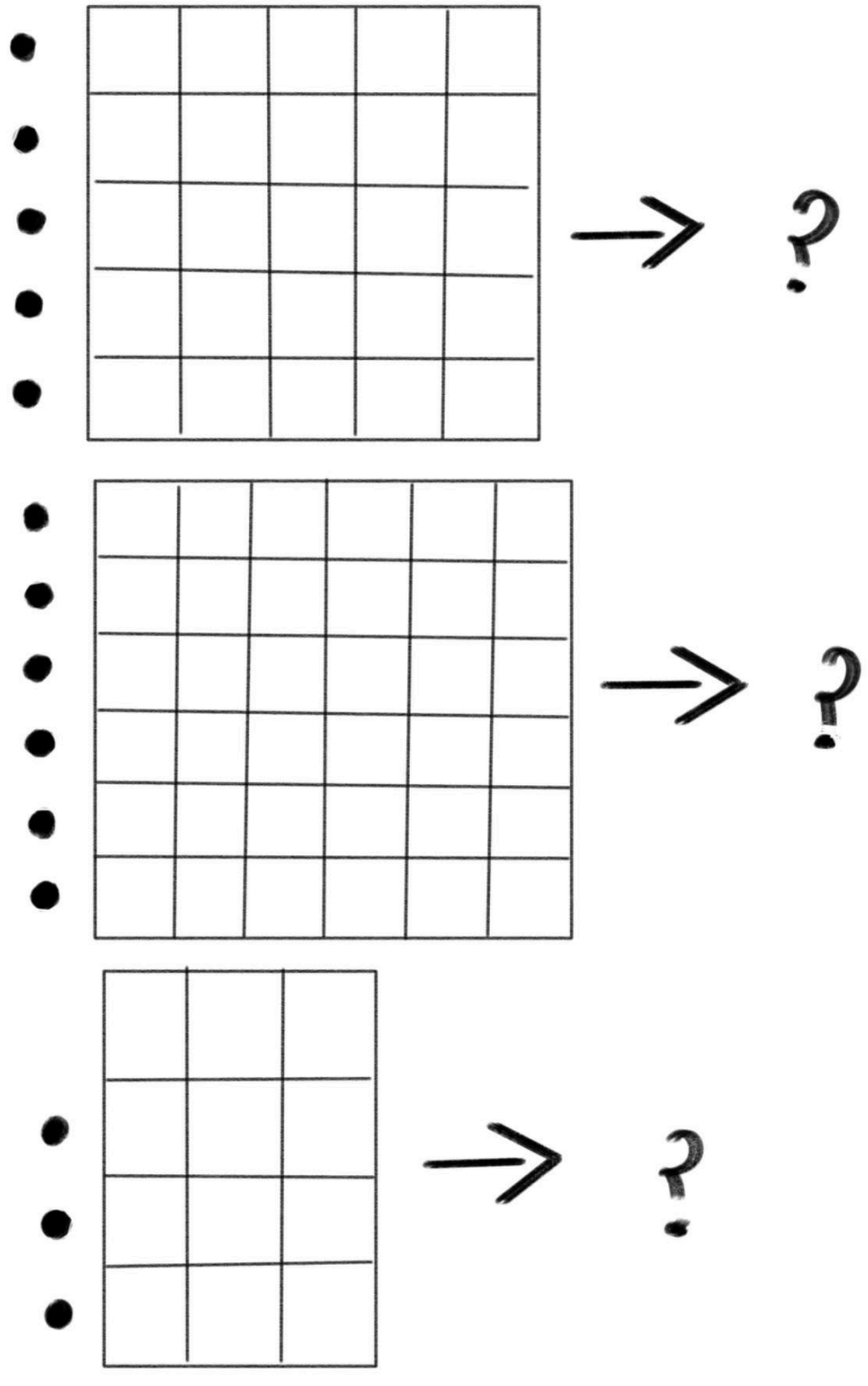
See and Solve Puzzle

Queen (Q) =



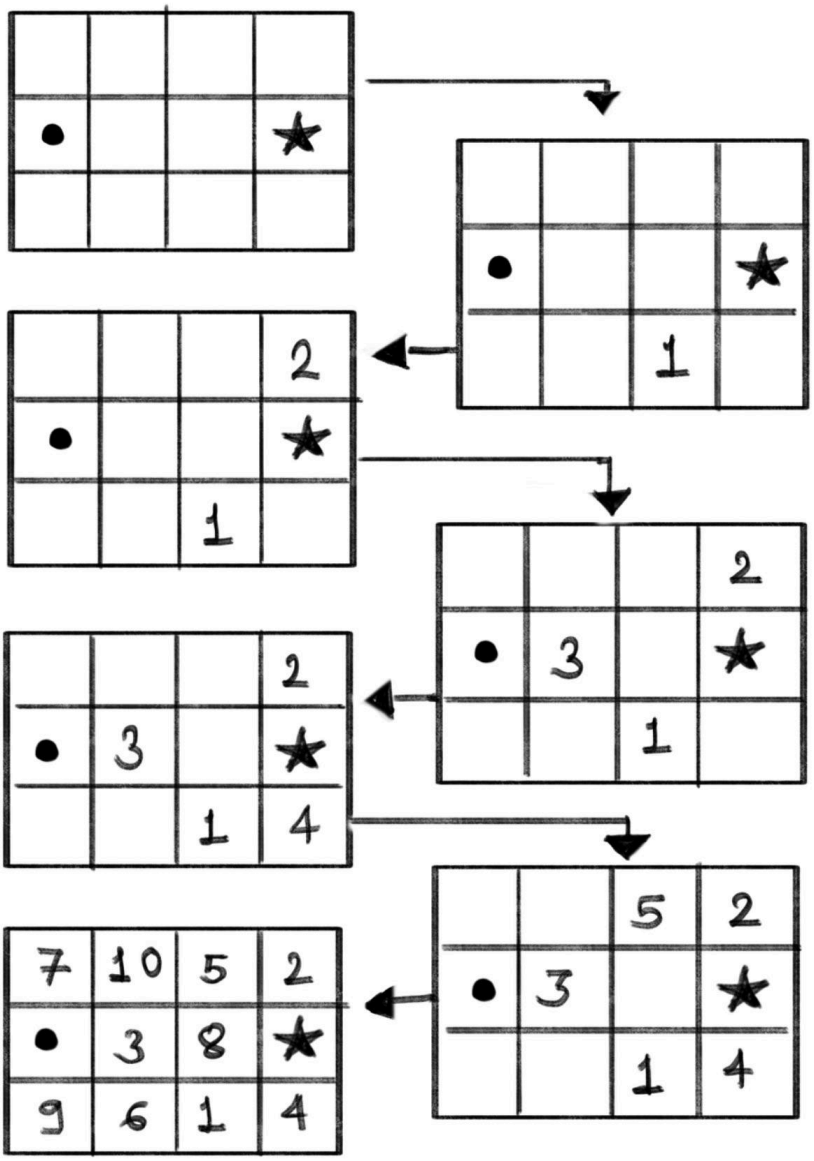


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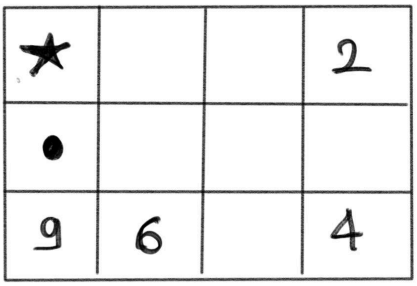


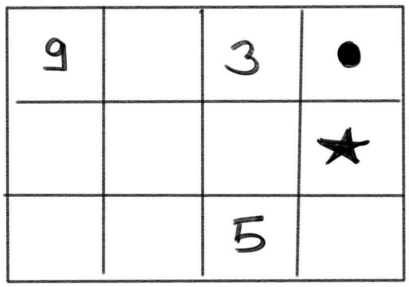
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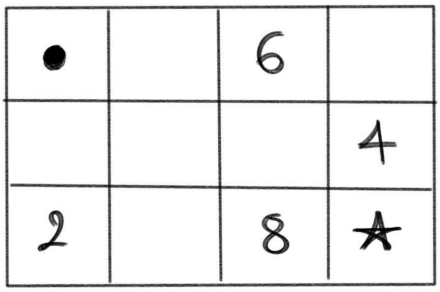


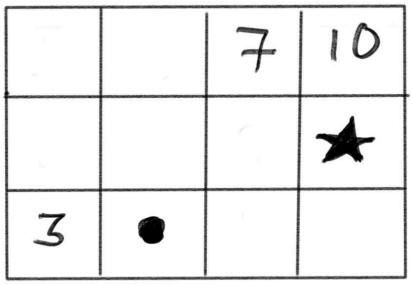


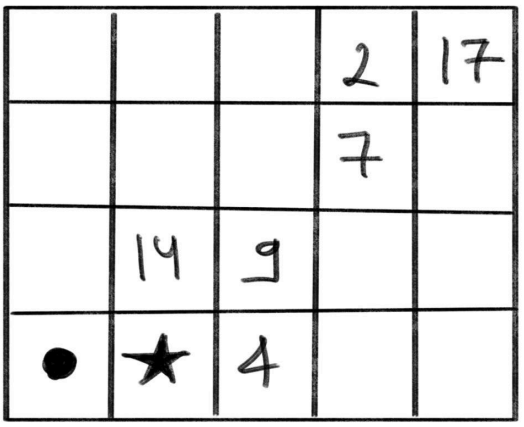
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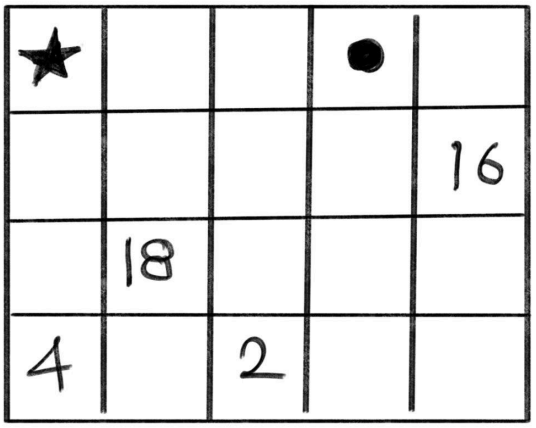












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